The 22 Minimal Topological Obstructions for a K_5 Minor

A complete description, explicit catalogue, and proof of minimality

Theorem. Let graphs be finite and simple. Up to isomorphism, there is a unique minimum set S such that a graph G contains K_5 as a minor if and only if G contains a subdivision of some graph in S. The set S has exactly 22 members.

1. Construction of the set

Label the vertices of K_5 by 1, 2, 3, 4, 5. At each vertex i, independently perform one of two operations: leave i unchanged; or split i into two adjacent vertices and partition its four neighbours into two pairs, one pair assigned to each new vertex. There are three possible 2+2 partitions at every vertex.

For each original edge ij, join the piece replacing i that receives neighbour j to the piece replacing j that receives neighbour i. Let S contain one representative of every isomorphism class obtained in this way.

Notation

An entry ab|cd in column i means that vertex i is replaced by adjacent vertices i' and i''; the first receives neighbours a,b, and the second receives neighbours c,d. A dash means that the vertex is not split.

number of vertices	5	6	7	8	9	10
isomorphism classes	1	1	2	6	6	6

2. The complete list of split systems

Graph	at 1	at 2	at 3	at 4	at 5
1	-	-	-	-	-
2	-	-	-	-	12|34
3	-	-	-	12|35	12|34
4	-	-	-	12|35	13|24
5	-	-	12|45	12|35	12|34
6	-	-	12|45	12|35	13|24
7	-	-	12|45	13|25	13|24
8	-	-	12|45	13|25	14|23
9	-	-	14|25	13|25	13|24
10	-	-	14|25	15|23	13|24
11	-	13|45	12|45	12|35	12|34
12	-	13|45	12|45	12|35	13|24
13	-	13|45	12|45	12|35	14|23
14	-	13|45	12|45	15|23	14|23
15	-	13|45	14|25	12|35	12|34
16	-	13|45	14|25	15|23	12|34
17	23|45	13|45	12|45	12|35	12|34
18	23|45	13|45	12|45	12|35	13|24
19	23|45	13|45	14|25	12|35	12|34
20	23|45	13|45	14|25	12|35	13|24
21	23|45	13|45	14|25	13|25	14|23
22	23|45	14|35	15|24	13|25	12|34

3. Why these graphs suffice

Suppose that G contains a K_5-minor. Choose a minor model. Retain one edge between each pair of its five branch sets, and inside each branch set retain a minimal tree connecting its four attachment points. Suppress every vertex of degree 2.

Consider one reduced branch tree T_i. Its vertices have total degree at least 3, where the total degree counts both tree edges and the four external edges. If $n_i = |V(T_i)|$, then the sum of these total degrees is

$$\mathbf{2(n_i - 1) + 4 = 2n_i + 2.}$$

Consequently $3n_i <= 2n_i + 2$, and therefore $n_i <= 2$. If $n_i = 1$, the corresponding vertex of K_5 is unsplit. If $n_i = 2$, equality holds throughout, so each end of the tree edge receives exactly two external edges. This is precisely a 2+2 split.

It follows that G contains a subdivision of one of the 22 graphs in the table. Conversely, contracting all split edges in any one of these graphs produces K_5. Thus a subdivision of any member has a K_5-minor.

4. Why all 22 graphs are necessary

Every graph H in S is connected and bridgeless. Splitting one vertex adds one vertex and one edge. Hence its cycle-space dimension is unchanged:

$$\mathbf{|E(H)| - |V(H)| + 1 = |E(K_5)| - |V(K_5)| + 1 = 10 - 5 + 1 = 6.}$$

A proper subgraph of a connected bridgeless graph has strictly smaller cycle-space dimension. Indeed, an omitted edge lies on a cycle, so the cycle space of the subgraph is a proper subspace of the original cycle space. Subdivision preserves cycle-space dimension. Therefore one member of S cannot contain a subdivision of a different member of S.

If any one of the 22 graphs were removed, taking G to be that omitted graph would disprove the equivalence. Consequently every member is necessary, and the 22-element set is the unique minimal set up to isomorphism.

5. The 22 graphs

Dark edges are inherited from K_5. Red edges are split edges; their colour is only a construction guide and is not part of the graph structure.